

Lab Activity: Dataset Ingestion and Integration from Multiple Structured Sources

Objective

To understand dataset ingestion from structured sources, tabular data representation, data quality assessment, and integration of multiple structured datasets using Python and Pandas.

Datasets Provided

1. students.csv
 2. attendance.xlsx
 3. placements.csv
 4. lms.json
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Part A: Dataset Ingestion

Question 1

Import the necessary Python libraries required for data ingestion and analysis.

Question 2

Load the Student Information dataset (students.csv) into a Pandas DataFrame.

Display:

- First five records
 - Last five records
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Question 3

Load the Attendance dataset (attendance.xlsx).

Display:

- First five records
 - Number of rows and columns
-

Question 4

Load the Placement dataset (placements.csv).

Display:

- Dataset shape
 - Column names
-

Question 5

Load the LMS dataset (lms.json).

Display:

- First five records
 - Data types of all attributes
-

Part B: Understanding Tabular Representation

Question 6

For each dataset, determine:

- Number of rows
 - Number of columns
 - Attribute names
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Question 7

Identify the common attribute that can be used to integrate all four datasets.

Question 8

Display the summary information of each dataset using suitable Pandas functions.

Question 9

Generate descriptive statistics for all numerical attributes.

Part C: Data Quality Assessment

Question 10

Check whether any dataset contains missing values.

Display the count of missing values for each attribute.

Question 11

Identify duplicate records in each dataset.

Display the number of duplicate records found.

Question 12

Determine whether the Student_ID column contains unique values in each dataset.

Question 13

Identify students who do not have LMS records.

Display their Student_ID values.

Part D: Data Integration

Question 14

Merge the Student dataset and Attendance dataset using the common key.

Display the resulting dataset.

Question 15

Integrate the Placement dataset with the merged dataset using a LEFT JOIN.

Display the resulting dataset.

Question 16

Integrate the LMS dataset with the existing dataset.

Display the final integrated dataset.

Question 17

Determine the total number of records and attributes in the final integrated dataset.

Question 18

Identify the attributes that contain missing values after integration.

Explain why these missing values occurred.

Part E: Data Transformation

Question 19

Replace missing values in the Company column with "Not Placed".

Question 20

Replace missing values in Salary_LPA with 0.

Question 21

Replace missing values in Videos_Watched and Assignments_Submitted with 0.

Question 22

Create a new attribute called LMS_Active.

Condition:

- True if Videos_Watched is greater than 0.
 - False otherwise.
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Part F: Data Analysis

Question 23

Display the student with the highest CGPA.

Question 24

Display the student with the highest attendance percentage.

Question 25

Compute the average CGPA of all students.

Question 26

Compute the average attendance percentage.

Question 27

Display department-wise average CGPA.

Question 28

Display department-wise average attendance.

Question 29

Count the total number of placed students.

Question 30

Find the company offering the highest salary package.

Question 31

Display students whose attendance is below 85%.

Question 32

Display students whose CGPA is greater than 8.5.

Question 33

Calculate the placement percentage.

Formula:

Placement Percentage = (Placed Students / Total Students) × 100

Question 34

Determine how many students actively use the LMS platform.

Part G: Reporting

Question 35

Create a bar chart showing department-wise average CGPA.

Question 36

Create a bar chart showing department-wise average attendance.

Question 37

Create a histogram showing attendance distribution.

Question 38

Create a pie chart showing placed vs. non-placed students.

Part H: Exporting Results

Question 39

Export the final integrated dataset to a CSV file named:

student_analytics.csv

Question 40

Save all generated visualizations as image files.